COURSE SYLLABUS

Administering a SQL Database Infrastructure 2025

COURSE NUMBER	DURATION	AUDIENCE LEVEL	FORMAT
55316B	**5 days · 15 modules**	**Intermediate**	**Instructor-Led / Self-Paced**

Course syllabus

COURSE OVERVIEW

This five-day instructor-led course provides students who manage SQL Server 2025 and Azure SQL databases with the knowledge and skills needed to administer a SQL Server 2025 database infrastructure. The material is also useful to individuals who develop applications that deliver content from SQL Server databases. In addition to SQL Server 2025 and Azure SQL Database, students will learn about SQL Server Management Studio 22 (SSMS 22).

AT COURSE COMPLETION

After completing this course, you will be able to:

- Authenticate and authorize users using Windows and SQL Server authentication modes, server and database roles, and granular permission management in SQL Server 2025 and Azure SQL Database
- Protect data using SQL Server Audit, temporal tables, Transparent Data Encryption, Always Encrypted, and Dynamic Data Masking in SQL Server 2025 and Azure SQL Database
- Design and implement comprehensive backup strategies and perform full, differential, transaction log, and advanced restore operations including point-in-time recovery in SQL Server 2025
- Automate database management using SQL Server Agent jobs, schedules, multi-server management, alerts, notifications, Database Mail, and PowerShell with the SqlServer module
- Monitor, trace, and troubleshoot SQL Server 2025 infrastructure using Dynamic Management Views, Extended Events, Performance Monitor, Management Data Warehouse reports, and structured troubleshooting methodology, and transfer data using bcp, BULK INSERT, SSIS, and data-tier application packages

PREREQUISITES

- Experience using applications on Windows Servers

- Experience working with SQL Server or another relational database management system
- Some experience with command-line tools such as PowerShell

COURSE OUTLINE

MODULE 1 SQL Server Security

140 min

This module covers SQL Server 2025 authentication modes, login account types, cross-server authorization using registered and linked servers, and partially contained databases. Students learn to configure and manage access at the instance level using SSMS 22.

LESSONS

- **Lesson 1:** Authenticating Connections to SQL Server 2025
- **Lesson 2:** Authorizing Logins to Connect to Databases
- **Lesson 3:** Authorization Across Server Instances
- **Lesson 4:** Partially Contained Databases and Database-Level Authentication
- **Lesson 5:** New Security Features in SQL Server 2025

LAB EXERCISES

- Configuring Authentication and User Authorization in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe Windows Authentication mode and Mixed Mode Authentication in SQL Server 2025
- Create and configure SQL Server and Windows login accounts using SSMS 22 and T-SQL
- Authorize logins to connect to databases by mapping them to database user accounts
- Configure registered servers and linked servers to enable cross-instance authorization
- Create and configure partially contained databases with database-level user authentication

MODULE 2 Assigning Server and Database Roles

123 min

This module covers fixed and user-defined server roles, fixed and user-defined database roles, and application roles in SQL Server 2025. Students learn to assign permissions to groups of users at the server and database level using the principle of least privilege.

LESSONS

- **Lesson 1:** Working with Fixed Server Roles in SQL Server 2025
- **Lesson 2:** Creating and Managing User-Defined Server Roles
- **Lesson 3:** Working with Fixed Database Roles
- **Lesson 4:** Creating User-Defined Database Roles
- **Lesson 5:** Configuring and Using Application Roles

LAB EXERCISES

- Managing Server and Database Roles in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe server-scoped permissions and assign fixed server roles to SQL Server 2025 logins
- Create and manage user-defined server roles with the principle of least privilege
- Describe fixed database roles and manage database role membership using SSMS 22 and T-SQL
- Create user-defined database roles with customized GRANT and DENY permissions
- Configure and use application roles with sp_setapprole and sp_unsetapprole

MODULE 3 Authorizing Users to Access Resources

123 min

This module covers granular permission assignment on database objects, code execution authorization for stored procedures and functions, CLR assembly security, ownership chains, and schema-level permission management in SQL Server 2025.

LESSONS

- **Lesson 1:** Authorizing User Access to Database Objects
- **Lesson 2:** Securing Stored Procedures and User-Defined Functions
- **Lesson 3:** CLR Integration and Code Access Security
- **Lesson 4:** Managing Ownership Chains
- **Lesson 5:** Configuring Permissions at the Schema Level

LAB EXERCISES

- Configuring Database Security and Access Control

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Use GRANT, REVOKE, and DENY to control principal access to database securables
- Authorize users to execute stored procedures and user-defined functions using appropriate permissions
- Describe CLR integration code access security levels: SAFE, EXTERNAL_ACCESS, and UNSAFE
- Explain ownership chains and how they simplify permission management across objects
- Configure schema-level permissions and assign default schemas to simplify object name resolution

MODULE 4 Protecting Data with Encryption and Auditing

140 min

This module covers auditing options including temporal tables, triggers, and SQL Server Audit, along with data encryption using TDE, Always Encrypted, and Dynamic Data Masking in SQL Server 2025 and Azure SQL Database.

LESSONS

- **Lesson 1:** Auditing Options: Triggers, Temporal Tables, and Common Criteria
- **Lesson 2:** Implementing SQL Server Audit Using Extended Events
- **Lesson 3:** Managing SQL Server Audit Output and State
- **Lesson 4:** Transparent Data Encryption and Migrating Encrypted Databases
- **Lesson 5:** Always Encrypted, Dynamic Data Masking, and Azure Encryption Options

LAB EXERCISES

- Implementing TDE, SQL Server Audit, and Dynamic Data Masking

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Create and query system-versioned temporal tables for tracking data changes in SQL Server 2025
- Implement SQL Server Audit at the server and database levels using Extended Events infrastructure
- Manage and retrieve SQL Server Audit output from binary files and Windows Event Logs
- Implement Transparent Data Encryption using the SMK, DMK, server certificate, and DEK hierarchy
- Apply Dynamic Data Masking to protect sensitive columns and describe the Always Encrypted feature

MODULE 5 Recovery Models and Backup Strategies

123 min

This module covers SQL Server 2025 recovery models, transaction log architecture, backup media and device options, and the design of comprehensive backup strategies that meet RTO and RPO objectives.

LESSONS

- **Lesson 1:** Choosing Backup Strategies: RTO, RPO, and Media Options
- **Lesson 2:** SQL Server 2025 Transaction Logs and Write-Ahead Logging
- **Lesson 3:** SQL Server 2025 Recovery Models: Full, Bulk-Logged, and Simple
- **Lesson 4:** Planning Full and Differential Backup Strategies
- **Lesson 5:** Transaction Log, Partial, Tail-Log, and Copy-Only Backup Types

LAB EXERCISES

- Lab 5: Understanding SQL Server 2025 Recovery Models

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe the Full, Bulk-Logged, and Simple recovery models and their impact on transaction log behavior
- Explain write-ahead logging, ACID properties, and log file truncation in SQL Server 2025
- Select appropriate backup media including local disk, network UNC paths, and Azure Blob Storage
- Design backup strategies combining full, differential, log, partial, and copy-only backups
- Define RTO and RPO objectives and align backup retention policies to meet them

MODULE 6 Backing Up SQL Server 2025 Databases

150 min

This module covers performing full, differential, transaction log, copy-only, compressed, and encrypted backups using SSMS 22, T-SQL, and PowerShell, along with backup integrity verification and history monitoring.

LESSONS

- **Lesson 1:** Backing Up Databases and Transaction Logs in SQL Server 2025
- **Lesson 2:** Managing Backup Integrity and Verification
- **Lesson 3:** Backup Retention, Testing Policies, and Cloud Storage
- **Lesson 4:** Copy-Only Backups and Backup Compression
- **Lesson 5:** Encrypting SQL Server 2025 Backups

LAB EXERCISES

- Performing and Managing SQL Server 2025 Database Backups

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Perform full, differential, and transaction log backups using SSMS 22, T-SQL BACKUP statements, and PowerShell
- Verify backup integrity using RESTORE VERIFYONLY and WITH CHECKSUM options
- Manage backup history using the msdb dbo.backupset table and SSMS Backup and Restore Events report
- Perform copy-only backups that do not affect scheduled differential backup chains
- Implement backup compression and AES 256 encrypted backups using certificate-based encryption

MODULE 7 Restoring SQL Server 2025 Databases

150 min

This module covers the full restore process including redo and undo phases, restoring full and differential backups with NORECOVERY and RECOVERY options, advanced restore scenarios including encrypted backups and individual data pages, and point-in-time recovery using STOPAT and STOPATMARK.

LESSONS

- **Lesson 1:** Understanding the SQL Server 2025 Restore Process
- **Lesson 2:** Restoring Full, Differential, and Transaction Log Backups
- **Lesson 3:** Advanced Restore Scenarios: Files, Pages, and System Databases
- **Lesson 4:** Restoring Encrypted Backups Across Instances
- **Lesson 5:** Point-in-Time Recovery: STOPAT and STOPATMARK

LAB EXERCISES

- Performing Database Restores and Point-in-Time Recovery

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Explain the restore process including tail-log backup, redo phase, and undo phase
- Restore databases from full, differential, and transaction log backups using SSMS 22 and T-SQL
- Perform advanced restore operations including encrypted backup restores and individual data page restores
- Identify which system databases require backups and apply appropriate restore strategies
- Execute point-in-time recovery using the STOPAT and STOPATMARK restore options

MODULE 8 Automating SQL Server 2025 Management

150 min

This module covers automating SQL Server 2025 management tasks using SQL Server Agent, including job creation with T-SQL and SSIS steps, schedule configuration, job scripting, job history monitoring, troubleshooting failed jobs, and multi-server management with master and target servers.

LESSONS

- **Lesson 1:** Automating SQL Server 2025 Management with SQL Server Agent
- **Lesson 2:** Defining Jobs, Job Steps, and Schedules
- **Lesson 3:** Scripting SQL Server Agent Jobs from SSMS 22
- **Lesson 4:** Monitoring and Troubleshooting SQL Server Agent Jobs
- **Lesson 5:** Multi-Server Management with Master and Target Servers

LAB EXERCISES

- Configuring and Troubleshooting SQL Server Agent Jobs

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe the benefits of automation and configure SQL Server Agent jobs with multiple step types and schedules
- Script SQL Server Agent jobs from SSMS 22 tasks and generate job creation T-SQL scripts
- Monitor job execution history using Log File Viewer, Job Activity Monitor, and msdb system tables
- Troubleshoot failed SQL Server Agent jobs by analyzing job history, error logs, and service account status
- Configure multi-server management with master and target servers and implement Maintenance Plan Wizard tasks

MODULE 9 Configuring Security for SQL Server Agent

140 min

This module covers SQL Server Agent security architecture, including the Agent service account, SQL Agent msdb roles, job step security contexts, credential creation and management, and proxy account configuration for non-T-SQL job steps in SQL Server 2025.

LESSONS

- **Lesson 1:** Understanding SQL Server Agent Security Architecture
- **Lesson 2:** Assigning Security Contexts to SQL Server Agent Job Steps
- **Lesson 3:** Configuring and Managing SQL Server Credentials
- **Lesson 4:** Creating and Managing SQL Server Agent Proxy Accounts
- **Lesson 5:** Troubleshooting SQL Server Agent Security Issues

LAB EXERCISES

- Implementing Proxy Accounts and Credentials for SQL Server Agent Jobs

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe the SQL Server Agent service account requirements and the three SQL Agent msdb database roles
- Explain how security context is applied to T-SQL and non-T-SQL job steps in SQL Server 2025
- Create and manage SQL Server credentials using SSMS 22, T-SQL CREATE CREDENTIAL, and New-SqlCredential
- Create proxy accounts and link them to job step subsystems to authorize external resource access
- Troubleshoot SQL Server Agent security failures by reviewing job history and error logs

MODULE 10 Monitoring SQL Server 2025 with Alerts and Notifications

140 min

This module covers SQL Server 2025 error monitoring, Database Mail configuration, SQL Server Agent operators and alert creation, and Azure SQL Database alert configuration using the Azure Monitor alerting infrastructure.

LESSONS

- **Lesson 1:** Monitoring SQL Server 2025 Errors and Error Logs
- **Lesson 2:** Configuring Database Mail in SQL Server 2025
- **Lesson 3:** Creating Operators and Configuring SQL Server Agent Alerts
- **Lesson 4:** Testing and Validating Alert Notifications
- **Lesson 5:** Configuring Alerts in Azure SQL Database

LAB EXERCISES

- Configuring SQL Server Alerts and Database Mail Notifications

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe SQL Server error structure including severity levels, and configure and cycle the SQL Server error log
- Configure Database Mail profiles, accounts, and security using the SSMS 22 Configuration Wizard
- Create SQL Server Agent operators and configure event alerts and performance condition alerts
- Test alert notifications using error simulation and verify notification delivery via Database Mail
- Configure Azure SQL Database alert rules with metrics, thresholds, and notification actions in the Azure Portal

MODULE 11 Introduction to Managing SQL Server 2025 by Using PowerShell 140 min

This module introduces the SqlServer PowerShell module, SQL Server Management Objects (SMOs), and PowerShell providers for navigating SQL Server 2025 resources. Students learn to configure databases and instances, manage logins and backups, and create Azure SQL Databases using PowerShell.

LESSONS

- **Lesson 1:** Getting Started with the SqlServer PowerShell Module
- **Lesson 2:** SQL Server Management Objects and Instance Navigation
- **Lesson 3:** Configuring SQL Server 2025 Database and Instance Settings with PowerShell
- **Lesson 4:** Managing Logins, Users, and Role Memberships with PowerShell
- **Lesson 5:** Backup, Restore, and Azure SQL Database Management with PowerShell

LAB EXERCISES

- Managing SQL Server 2025 Databases and Logins with PowerShell

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe the SqlServer PowerShell module, SMO namespaces, and PowerShell providers for SQL Server 2025
- Read and configure SQL Server 2025 database settings using Get-SqlDatabase and SMO object methods
- Configure SQL Server 2025 instance settings using Get-SqlInstance and SMO methods with service restart
- Manage SQL Server 2025 logins, users, and role memberships using PowerShell cmdlets and Invoke-Sqlcmd
- Perform database backup and restore operations and manage Azure SQL Databases using Az.Sql cmdlets

MODULE 12 Tracing Access to SQL Server 2025 with Extended Events

105 min

This module covers the Extended Events monitoring framework in SQL Server 2025 including sessions, events, actions, targets, and predicates. Students learn to create and manage sessions using T-SQL and SSMS 22 and apply best practices for performance-aware event monitoring.

LESSONS

- **Lesson 1:** Extended Events Core Concepts: Sessions, Events, and Actions
- **Lesson 2:** Creating Extended Events Sessions Using T-SQL and SSMS 22
- **Lesson 3:** Working with Extended Events Live Data and XEL Output
- **Lesson 4:** Extended Events Best Practices and Use Cases

LAB EXERCISES

- Monitoring Query Performance and Blocking with Extended Events

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe Extended Events core concepts including sessions, events, actions, targets, predicates, and packages
- Create Extended Events sessions using T-SQL CREATE EVENT SESSION and the SSMS 22 New Session Wizard
- Query and analyze live Extended Events data and XEL file output in SSMS 22
- Apply Extended Events best practices including using the system_health session and event filtering
- Identify use cases for Extended Events including lock and block analysis, query execution monitoring, and system health

MODULE 13 Monitoring SQL Server 2025

140 min

This module covers Dynamic Management Views, Activity Monitor, Performance Monitor with SQL Server counters, the Management Data Warehouse, System Data Collection Sets, and SSMS reports for analyzing disk usage, query statistics, and server activity in SQL Server 2025.

LESSONS

- **Lesson 1:** Using Dynamic Management Views and Functions in SQL Server 2025
- **Lesson 2:** Activity Monitor and Real-Time Instance Monitoring in SSMS 22
- **Lesson 3:** Performance Monitor and SQL Server 2025 Counters
- **Lesson 4:** Configuring the Management Data Warehouse and Data Collection Sets
- **Lesson 5:** Analyzing Management Data Warehouse Reports

LAB EXERCISES

- Monitoring SQL Server 2025 Performance and Activity

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Query Dynamic Management Views and Functions to monitor SQL Server 2025 sessions, I/O, and performance counters
- Use Activity Monitor in SSMS 22 to review real-time process, I/O, resource wait, and query data
- Add SQL Server 2025 performance counters to Performance Monitor and configure Data Collector Sets
- Configure the Management Data Warehouse and System Data Collection Sets using the SSMS 22 wizard
- Analyze Disk Usage Summary, Query Statistics History, and Server Activity History reports from the management data warehouse

MODULE 14 Troubleshooting SQL Server 2025

123 min

This module covers a structured troubleshooting methodology for SQL Server 2025 and applies it to service-related issues, connectivity problems, and login failures. Students use SQL Server error logs, Windows Event Logs, and SSMS 22 to diagnose and resolve common infrastructure problems.

LESSONS

- **Lesson 1:** Applying a SQL Server 2025 Troubleshooting Methodology
- **Lesson 2:** Resolving SQL Server 2025 Service-Related Issues
- **Lesson 3:** Troubleshooting SQL Server 2025 Connectivity Issues
- **Lesson 4:** Troubleshooting Login and Authentication Failures in SQL Server 2025
- **Lesson 5:** Using Windows Event Logs and SSMS 22 for Efficient Diagnostics

LAB EXERCISES

- Diagnosing and Resolving SQL Server 2025 Service and Connectivity Issues

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Apply a four-phase troubleshooting methodology (Investigate, Analyze, Implement, Validate) to SQL Server 2025 problems
- Resolve SQL Server 2025 service startup failures using SQL Server error logs and text editors when SSMS is unavailable
- Troubleshoot SQL Server 2025 connectivity issues by testing Shared Memory, Named Pipes, TCP/IP, and firewall settings
- Diagnose and resolve Windows login, SQL Server login, and contained database user authentication failures
- Use Windows Event Log filtering and SQL Server error log cycling to efficiently isolate and document issues

MODULE 15 **Importing and Exporting Data** 213 min

This module covers ETL and ELT data transfer patterns, constraint management for bulk operations, SSIS and the Import and Export Wizard, bcp and BULK INSERT command usage, OPENROWSET, and deployment of database applications using DACPAC and BACPAC files in SQL Server 2025.

LESSONS

- **Lesson 1:** ETL, ELT, and SQL Server 2025 Data Transfer Tools
- **Lesson 2:** Disabling and Enabling Constraints for Bulk Import Operations
- **Lesson 3:** Using Linked Servers and the Import and Export Wizard
- **Lesson 4:** Importing Data with bcp and BULK INSERT
- **Lesson 5:** Deploying Database Applications with DACPAC and BACPAC

LAB EXERCISES

- Importing and Exporting Data with Multiple Methods

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Describe ETL and ELT data transfer patterns and select appropriate SQL Server 2025 tools for each scenario
- Disable and re-enable table constraints and indexes to optimize bulk import performance
- Transfer data between SQL Server 2025 instances using the SSMS 22 Import and Export Wizard and SSIS packages
- Import data using the bcp command-line utility and T-SQL BULK INSERT with custom format, delimiter, and row terminator options
- Create and deploy DACPAC and BACPAC files using SSMS 22 wizards for SQL Server 2025 and Azure SQL Database

BOOTCAMP MODE — CAPSTONE LABS

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3

SKILLS TO BE INTEGRATED

- Configure SQL Server 2025 authentication modes and create SQL Server login accounts using SSMS 22
- Map login accounts to database users and assign default schemas in SQL Server 2025
- Create and configure registered servers and linked servers across SQL Server 2025 instances
- Create and test partially contained databases with database-level user authentication
- Assign fixed and user-defined server-level roles to SQL Server 2025 logins using T-SQL and SSMS 22
- Create fixed and user-defined database roles and manage role membership in SQL Server 2025
- Configure application roles using sp_setapprole and sp_unsetapprole
- Grant, deny, and revoke object-level permissions on tables, views, and stored procedures
- Authorize users to execute stored procedures and scalar and table-valued functions
- Configure schema-level permissions and assign default schemas to database users

DAY 2 CAPSTONE Modules 4, 5, 6

SKILLS TO BE INTEGRATED

- Create and query system-versioned temporal tables for auditing data changes in SQL Server 2025
- Implement SQL Server Audit at the server and database level using SSMS 22 and T-SQL
- Manage and view SQL Server Audit output from binary files and Windows Event Logs
- Configure Dynamic Data Masking on sensitive columns using default, email, custom, and random mask formats
- Implement Transparent Data Encryption including the SMK, DMK, server certificate, and DEK hierarchy
- Configure full, differential, and transaction log backups using SSMS 22, T-SQL, and Azure Blob Storage
- Configure backup compression, copy-only backups, and encrypted backups with AES 256
- Verify backup integrity using RESTORE VERIFYONLY and WITH CHECKSUM
- Plan and implement backup retention policies including testing and multi-location strategies
- Configure SQL Server 2025 recovery models (Full, Bulk-Logged, Simple) and evaluate transaction log capacity

DAY 3 CAPSTONE Modules 7, 8, 9

SKILLS TO BE INTEGRATED

- Perform full, differential, and transaction log restores using SSMS 22, T-SQL, and PowerShell
- Execute point-in-time recovery using STOPAT and STOPATMARK restore options
- Restore encrypted backups by migrating certificates between SQL Server 2025 instances
- Restore individual corrupted data pages using full backup and log sequences
- Create and manage SQL Server Agent jobs with T-SQL, PowerShell, and CmdExec job step types
- Configure job schedules, script jobs to files, and manage job history using msdb system tables
- Configure multi-server management with master and target servers and Maintenance Plan Wizard
- Configure the SQL Server Agent service account and assign SQLAgentUserRole, SQLAgentReaderRole, and SQLAgentOperatorRole
- Create and manage SQL Server credentials using SSMS 22, T-SQL CREATE CREDENTIAL, and New-SqlCredential
- Create and assign proxy accounts to non-T-SQL job step subsystems including PowerShell and SSIS

DAY 4 CAPSTONE Modules 10, 11, 12

SKILLS TO BE INTEGRATED

- Configure Database Mail profiles, accounts, and SMTP settings using the SSMS 22 Configuration Wizard
- Create SQL Server Agent operators and configure SQL Server event alerts tied to specific error numbers and severity levels
- Configure Azure SQL Database alert rules using the Azure Portal with metric thresholds and notification actions
- Install and import the SqlServer PowerShell module and navigate SQL Server objects using the SQLSERVER provider
- Configure SQL Server 2025 database and instance settings using SMOs, Get-SqlDatabase, and Get-SqlInstance
- Manage SQL Server 2025 logins, users, and role memberships using Add-SqlLogin, Add-RoleMember, and Invoke-Sqlcmd
- Perform SQL Server 2025 database backup and restore operations using Backup-SqlDatabase and Restore-SqlDatabase
- Create and manage Azure SQL Databases using New-AzResourceGroup, New-AzSqlServer, and New-AzSqlDatabase
- Create and configure Extended Events sessions using T-SQL and the SSMS 22 New Session Wizard
- Query and analyze Extended Events live data and XEL file output filtered by database and event type

DAY 5 CAPSTONE Modules 13, 14, 15

SKILLS TO BE INTEGRATED

- Monitor SQL Server 2025 instance activity using Dynamic Management Views, Activity Monitor, and Performance Monitor with SQL Server counters
- Configure the Management Data Warehouse and System Data Collection Sets using the SSMS 22 wizard
- Analyze Disk Usage Summary, Query Statistics History, and Server Activity History reports from the management data warehouse
- Apply a structured troubleshooting methodology (Investigate, Analyze, Implement, Validate) to SQL Server 2025 errors
- Resolve SQL Server 2025 service-related issues using SQL Server error logs and Windows Event Logs
- Troubleshoot SQL Server 2025 connectivity and login issues for Windows logins, SQL Server logins, and contained database users
- Disable and re-enable table constraints and indexes to optimize bulk import operations
- Transfer data between SQL Server 2025 instances using the SQL Server Import and Export Wizard and SSIS packages
- Import data using bcp command-line utility and BULK INSERT T-SQL statement with custom format and delimiter options
- Create and deploy DACPAC and BACPAC files using SSMS 22 Extract and Export Data-tier Application wizards

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.